

Faux Trading — Project Presentation Script

Opening Statement

"Good morning/afternoon, Sir/Ma'am. I'd like to present our capstone project — Faux Trading, an AI-powered virtual stock trading simulator built specifically for the Indian market."

"The core idea is simple: people want to learn trading but are afraid of losing real money. Faux Trading solves that by giving every user ₹84 lakh in virtual funds to trade 10 real Indian stocks — with zero financial risk."

1. Landing Page (/)

"This is our landing page. Let me walk you through what it communicates."

- The headline: **"Master markets. Zero risk."** — instantly tells the user what the product does
- Below that, three key stats: ₹84 lakh virtual funds, 20+ assets, ₹0 real money needed
- A live product preview on the right shows an actual portfolio dashboard so users know exactly what they're getting
- The **ticker strip** at the top shows live-simulated Indian stock prices — SBIN, TCS, RELIANCE, INFY — scrolling in real time
- We have a "How it works" section explaining the 5-step journey from signup to trading
- A features section covering all 8 capabilities
- A leaderboard preview showing risk-adjusted rankings
- And a strong CTA at the bottom

"The design is intentionally minimal and premium — we did not want this to look like a generic AI-generated website. Every component was deliberately designed."

2. Authentication (/login, /register)

"Authentication is standard but polished."

- Register: user enters name, username, email, password
 - On registration, they get ₹84,00,000 in virtual funds immediately — no payment, no card
 - Login authenticates and redirects straight to the dashboard
 - The register page clearly states "virtual funds only, no real money" — this is an important legal and ethical disclaimer
 - Sessions are managed via JWT tokens on the backend
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3. Dashboard (/dashboard)

"After login, the user lands here. This is the central hub."

Four stat cards at the top:

- Portfolio Value — total worth of cash + holdings
- Total Return — how much they've made or lost since starting
- Unrealized P&L — profit/loss on open positions
- Virtual Cash — available to invest right now

Main content:

- A **90-day performance chart** showing portfolio growth vs invested amount
- A **holdings table** showing every stock they own, with current price, today's change, market value, P&L, and allocation percentage
- Recent activity showing their last 4 trades

Right sidebar:

- Live NIFTY, SENSEX, Bank Nifty, IT Index, India VIX snapshots
- Watchlist with 6 stocks they're tracking
- An AI insight card showing a portfolio-level analysis
- Quick trade button that takes them directly to SBIN's trading page

"The dashboard is designed around information hierarchy — the most important number, portfolio value, is the first thing you see. Everything else supports it."

4. Market Page (/market)

"This is where users explore all 10 Indian stocks."

- At the top, live index cards for NIFTY 50, SENSEX, Bank Nifty, Nifty IT, India VIX, and USD/INR
- Below that, **Top Gainers** and **Top Losers** sections — at a glance, users know what's moving today
- A searchable, filterable table of all 10 stocks with:
 - Current price in ₹
 - Today's change (% and absolute)
 - Volume
 - Market cap
 - 52-week range visualised as a bar chart
 - Watchlist button to track a stock

"Sector filters let users quickly narrow down to Financial, Technology, Energy, Industrials etc."

5. Asset Detail Page (/market/SBIN etc.)

"When you click any stock, you get a full detail page. This is one of our strongest pages."

Header: Stock symbol, company name, exchange (NSE), current price, and today's change with a directional arrow

Price Chart:

- Built using Lightweight Charts library — the same library used by TradingView
- Supports Candlestick and Line view
- Time ranges: 1D, 1W, 1M, 3M, 6M, 1Y, 5Y
- Crosshair shows exact price at any point

Key Statistics panel: Open, High, Low, Previous Close, Volume, Market Cap, P/E, EPS, 52-week range

XGBoost AI Prediction panel — this is where our ML model integrates:

- Direction: BULLISH / BEARISH / NEUTRAL
- Model accuracy (e.g., TCS: 82.04%)
- ROC-AUC score
- Probability bar: e.g., TCS shows 15.3% UP, 84.7% DOWN
- 5-day price estimate
- Clear disclaimer that this is not investment advice

Trade Panel on the right:

- Shows current position if the user already holds this stock
- Buy/Sell toggle
- Market or Limit order type
- Quantity input with +/- controls
- Live fee calculation (0.1% simulated brokerage)
- Balance check — shows error if insufficient funds
- Confirmation step before executing — prevents accidental trades

"We deliberately added a confirmation screen. In real trading apps, accidental trades can cost thousands. We wanted to teach that discipline."

6. Portfolio Page (/portfolio)

"This gives the user a complete picture of their investment performance."

Six metric cards:

- Total Value, Total Return, Unrealized P&L, Realized P&L, Cash, Invested Amount

Two charts:

- 90-day performance line chart — portfolio value vs invested amount
- Donut allocation chart — how portfolio is distributed across stocks

Full holdings table with every position — quantity, average cost, current price, market value, day P&L, total P&L, and allocation percentage

"Everything here is calculated server-side. The frontend never computes balances — it only displays what the backend returns. This prevents any manipulation."

7. Orders Page (/orders)

"This shows the complete order history."

- Tab filters: All, Open, Filled, Cancelled
 - Table shows symbol, side (buy/sell), order type (market/limit), quantity, limit price, fill price, total value, status, and timestamp
 - Open orders have a Cancel button
 - Status badges: Filled (green), Open (blue), Cancelled (grey), Rejected (red)
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8. Transactions Page (/transactions)

"This is the confirmed trade history — only filled orders appear here."

- Shows every executed trade with exact fill price, quantity, fees charged, total cost
 - For sell transactions, shows realized P&L — how much profit or loss was made on that trade
 - Useful for performance review and tax purposes in a real scenario
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9. Strategies Page (/strategies)

"This is where the educational power of the app really shows."

Strategy Builder:

- Users create rule-based trading strategies using a visual interface — no coding required
- Each rule follows the format: **WHEN [indicator] [condition] [value] THEN [BUY/SELL] with [X% of portfolio]**
- Example: "WHEN 50-day SMA crosses above 200-day SMA THEN BUY with 100% of portfolio" — this is the classic Golden Cross strategy
- Supported indicators: SMA 20/50/200, EMA 12/26, RSI 14, MACD, Bollinger Bands, Volume

Pre-built strategies:

- Golden Cross — for SBIN (SMA crossover)
- RSI Bounce — for TCS (mean reversion)
- Each shows performance metrics: return, Sharpe ratio, max drawdown, win rate

"We intentionally kept strategy creation structured rather than allowing arbitrary code. This is safer and more educational for non-technical users."

10. Backtesting Page (/backtesting)

"This is one of the most technically impressive features."

- User selects a strategy, stock, date range, starting capital, fees, slippage
- Clicks Run Backtest — the engine simulates the strategy against historical price data
- **Results include 12 performance metrics:**
 - Total Return and Benchmark Return (vs NIFTY 50)
 - CAGR (Compound Annual Growth Rate)
 - Sharpe Ratio and Sortino Ratio
 - Maximum Drawdown and Duration
 - Win Rate, Profit Factor
 - Total Trades, Average Trade Return, Volatility
- **Portfolio Value Chart** — shows strategy performance vs benchmark over the full period
- **Trade History Table** — every simulated buy/sell with price, quantity, value, and P&L

"Users can see concretely whether their strategy would have worked historically before risking any virtual money. This teaches the discipline of testing before trading."

11. AI Insights Page (/insights)

"This page is powered by the XGBoost model built by our ML team."

Model overview banner:

- Global XGBoost model — one shared model for all 10 companies
- Average accuracy: 81.3%
- Average ROC-AUC: 0.890
- Prediction horizon: 5 trading days

Summary cards: Count of bullish, bearish, and neutral signals across all 10 stocks

10 prediction cards, one per stock, showing:

- Direction badge: BULLISH / BEARISH / NEUTRAL
- Current price
- UP/DOWN probability bar — e.g., Infosys shows 16.7% UP, 83.3% DOWN
- Confidence level
- 5-day price estimate
- Model accuracy for that specific stock

Current predictions from the model:

- SBI → UP (71.1% probability)
- Aditya Birla Capital → UP (65%)
- Larsen & Toubro → UP (66.7%)
- ICICI Bank → DOWN (74.6%)
- Infosys → DOWN (83.4%)
- TCS → DOWN (84.7%)

Model accuracy table showing accuracy, balanced accuracy, and ROC-AUC per company — TCS has the highest ROC-AUC at 0.914

"These are real predictions from the actual XGBoost model trained by our ML team. The frontend was built with clean interfaces so the real model plugs in without any redesign."

12. AI Coach Page (/coach)

"This is our conversational AI trading coach."

- Chat interface where users can ask any trading question
- The coach is not just a generic chatbot — it has context about the user's portfolio and recent trades
- **Quick questions on the right** for common topics: RSI, moving averages, position sizing, Sharpe ratio, portfolio risk
- Example: Ask "What is RSI?" and the coach explains it in plain language with actionable advice specific to Indian markets
- Ask "Analyze my portfolio" and it discusses sector concentration, risk exposure, and suggestions

"The design goal was: feel like a knowledgeable friend, not a formal financial advisor. It explains without prescribing."

13. Leaderboard Page (/leaderboard)

"Competition drives engagement and better trading habits."

Key design decision: Rankings are NOT based on raw returns. They're based on **risk-adjusted performance** — Sharpe ratio, win rate, and total return weighted equally.

- A trader with 35% return and disciplined risk management ranks higher than someone who made 50% by gambling on one trade
- Podium shows top 3 with gold/silver/bronze medals
- Full table shows rank, username, portfolio value, total return, Sharpe ratio, win rate, trade count
- Current user is highlighted in amber

"This teaches users that good trading is about consistency and risk management, not just making big gains. Someone who takes 10x leverage to make 100% in a day is not a good trader."

14. Learn Page (/learn)

"Structured learning to build trading knowledge progressively."

5 modules:

1. Market Basics — 6 lessons, 20 min
 2. Reading Charts — 6 lessons, 30 min
 3. Risk Management — 6 lessons, 25 min
 4. Strategies — 6 lessons, 35 min (unlocks after basics)
 5. Portfolio Building — 5 lessons, 20 min (unlocks after strategies)
- Progress tracking — circular progress indicator shows overall completion percentage
 - Each lesson opens as a modal with plain-language content, practical examples using Indian stocks
 - "Mark as done" button tracks progress
 - Continue button always shows the next pending lesson

"The learning center is integrated with the rest of the app — lessons reference the chart page, backtesting, and AI coach directly."

15. Profile Page (/profile)

"Standard user settings page."

- Editable display name, username, email — changes reflect immediately in the sidebar
- Performance stats: portfolio value, total return, Sharpe ratio, win rate
- 4 achievements: First Profit, Strategist, Backtester, Top 10

- Notification preferences with toggle switches
 - Password change
 - Portfolio reset option — resets virtual balance back to ₹84 lakh
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Technical Architecture Summary

"Let me quickly summarise the technical stack."

Frontend:

- Next.js 16 with TypeScript — server-side rendering, static generation
- Tailwind CSS with custom design tokens — our own colour palette, not default Tailwind
- Lightweight Charts for financial price charts (same library as TradingView)
- Recharts for portfolio and performance charts
- Zustand for state management

Backend:

- FastAPI with Python — high performance, async
- SQLAlchemy + PostgreSQL — all financial data stored server-side
- All balance and position calculations happen on the server — the frontend is never trusted
- JWT authentication
- Redis for caching

AI/ML:

- XGBoost global model by our ML team — one model for all 10 Indian stocks
- Clean service interface: MockAIService → XGBoostAIService → future MLModelAIService
- The frontend doesn't need to change when the model is upgraded

Security:

- Server-side validation on every trade — quantity, balance, position checks all enforced server-side
- No client-side trust for any financial calculation
- Database-level constraints prevent invalid data

Testing:

- 20 backend tests covering trading engine, authentication, backtesting, AI service
 - All 20 passing
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Closing Statement

"To summarise: Faux Trading is a complete, production-quality fintech simulator. It covers the full trading lifecycle — from learning to execution to performance review — while being safe, educational, and engaging. The XGBoost model from our ML team integrates cleanly without requiring any frontend changes. And the entire app is built on real financial concepts — no shortcuts, no placeholder logic where it matters."

"Thank you. I'm happy to take questions or demonstrate any specific feature."